supplier since 1999, accounting for an estimated 70–90% of supply [49, 58]. Apart from exerting a large influence on global drug markets, the cultivation of opium poppy has had significant implications on Afghanistan's economic development, security situation, and rural communities. In 2019, opium cultivation generated an estimated income of $1.2–$2.1 billion domestically, or around 10% of Afghanistan's gross domestic product [55]. Villages that cultivated opium poppy have been found (in surveys) to have poorer access to infrastructure and basic services, and experience more violence [55]. Yet few avenues exist to generate more comprehensive data and evidence on opium poppy cultivation and its associated local socioeconomic conditions, without the use of costly surveys or data collection efforts.

Over the years, efforts to monitor opium poppy cultivation in Afghanistan have been led by the United Nations Office of Drugs and Crime (UNODC). The UNODC releases annual reports containing aggregated estimates of opium cultivation, both at a province and at a district level (Afghanistan is divided into 32 provinces and 398 districts; district-level UNODC estimates are described as being "indicative") [7, 49, 55]. The methodology involves field surveys and manual annotation of high-resolution (e.g., <1 m) satellite imagery. This approach produces valuable information, but is expensive and time-consuming, and difficult to undertake under poor security conditions. The Afghanistan government had previously been accused of blocking publication of poppy cultivation-related statistics [9]; in some years, reports are released with long delays.[1] In the past, the UNODC has additionally conducted separate surveys among village headmen in certain years to gather village-level information about poppy cultivation and socioeconomic conditions [55]. Given the current political environment, the future availability or reliability of official estimates or surveys is unclear.[2]

The focus of this work is to produce high-resolution maps of opium poppy cultivation, covering a large geographical area, using only freely available satellite imagery. Apart from cost and availability, there are numerous arguments for the production and use of such high-resolution maps, relative to aggregate statistics. A primary one is that many phenomena are local, and analyses conducted at an aggregated level obscure local patterns. With the public availability of high-resolution satellite imagery and other sources of granular data, there has been a proliferation of related research producing grid-level maps in numerous domains, for example, for economic development and public health-related outcomes (e.g., population [52], poverty [26], electrification [17], infant, and child mortality [14]). The availability of high-resolution cultivation maps will allow us to exploit these rich new sources of data to gain additional insight on local circumstances surrounding cultivation. Specific to the Afghan context, maps can overcome past over-reliance on aggregated cultivation statistics—previous counter-narcotics efforts have focused heavily on reducing overall opium cultivation area, which has been criticized as being counterproductive, because they do not take into consideration local socioeconomic challenges and the availability of alternative livelihoods [49]. Given the challenging political and security situation after the Taliban takeover, generating maps from freely-available data could be a valuable additional source of information on local conditions moving forward.

Mapping opium poppy relates more broadly to the rich literature on land cover classification as well as crop type mapping. However, existing methods do not transfer easily, since only major crops like wheat, maize, rice, and soybean are typically considered. While there has been some existing work using automated methods to detect the opium poppy crop using remotely sensed

[1]For example, the report on the 2019 harvest season was only released in February 2021.
[2]Since the Taliban takeover in August 2021, a full report of 2021 cultivation statistics was released; as of May 2023, province-level cultivation statistics for the 2022 season have been released. No findings from socioeconomic surveys have been released to our knowledge.

often generated from imagery acquired at multiple time points throughout the year [18, 24]. Vegetation indices (such as the normalized difference vegetation index (NDVI)) are calculated at each time point, capturing information about crop cycles which can be used to differentiate between crops. Apart from using spectral properties at multiple time points directly, Fourier transforms have been used to derive features from time series profiles of both surface reflectance and vegetation indices [37, 62]. For classification, both unsupervised and supervised [e.g., 42] methods have been employed, for example, *k*-means, Gaussian mixture models (GMM) [e.g., 62], random forests, and deep learning approaches [e.g., 27]. For the identification of single crops, acquisition windows (for acquiring satellite imagery) may be timed for the optimal detection of that particular crop, either from knowledge of crop calendars or measured from the imagery (using indices such as the NDVI) [53]. For example, suitable acquisition timing may be a pre-harvest or post-harvest period, or multiple months corresponding to the phenological stages of the crop [8]. Such a strategy avoids using imagery from time periods which do not provide useful information about the particular crop. Spectral properties or vegetation indices are then used in a similar manner to generate features for unsupervised or supervised methods.

2.2 Automatically Detecting Opium Poppy

More specific to our problem, researchers have tested methods to specifically detect the opium poppy crop at a field-level. This has been done using satellite imagery as well as imagery from unmanned aerial vehicles and field spectrometers [16, 28, 31, 32, 45, 60, 61, 67]. A comprehensive review on detecting cannabis and opium crops is in [8]. Due to the illegal nature of the opium poppy crop, existing work has primarily been done under experimental settings [e.g., 28], or in collaboration with drug monitoring agencies that have access to ground truth data in a limited geographical area of interest [e.g., 60, 61]. Many of these studies are in Afghanistan, since majority of the world's opium cultivation occurs in the country. The processing and classification steps largely mirror those for crop type mapping for single crops; typical pre-processing of imagery includes atmospheric correction and the removal of cloudy pixels.

In terms of image acquisition, satellite imagery is first acquired during the appropriate period in the crop cycle, in particular pre-harvest, close to the peak in vegetation activity and when crops are flowering [28, 47, 53]. This is either done in experimental settings or guided by knowledge of crop calendars in poppy-growing areas. For supervised approaches, labels are obtained either through experimental settings, visual interpretation of known cultivation areas, or in collaboration with the UNODC. Specific methods that have been tested include classification and regression trees [28], maximum-likelihood [16], nearest neighbors [46], as well as deep learning approaches for object detection [31, 67], and semantic segmentation [32]. For unsupervised approaches, methods like Iterative Self-Organizing Data Analysis Technique (ISODATA), a variant of the standard k-means algorithm, have been used, as well as more specialized algorithms such as Multiple Endmember Spectral Mixture Analysis (MEMSA) [45, 60, 61]. Our analysis focuses on unsupervised methods, since these do not require manual annotation or access to previously constructed ground truth data sets. Ground truth data require expert knowledge and/or verification using ground images, which are not widely accessible or feasible.

While not an automated strategy, it is worth briefly reviewing the methodology for the production of official statistics released by the UNODC and the U.S. government in Afghanistan [7, 49, 55], since these are (to our knowledge) the only source of data covering a large geographical area, although at an aggregated level. In high-cultivation districts, high-resolution (e.g., <1 m) satellite imagery is acquired for sampled locations at a pre-harvest and post-harvest date. These are manually annotated; annotations from these sampled locations are then used to produce aggregated estimates, incorporating information about the sampling strategy. The sampling strategy is

Table 1. Distinguishing Characteristics of Wheat Versus Poppy

Crop	Harvest dates	Crop characteristics	
Poppy	April–June [53, 54]	Pre-harvest	NDVI: slowly increasing to a peak of around 0.6 [54]
			Appearance: Bright green [7, 31]
		Post-harvest	Fields often plowed immediately after harvest [54]
			Appearance: Light brown / beige [7]
Wheat	Typically right after poppy [22]	Pre-harvest	NDVI: consistently hovering around 0.8 [54]
			Appearance: Dark green [31]
		Post-harvest	Not plowed as quickly [54]
			Appearance: Dark brown

In many districts, particularly those with larger amounts of cultivation, these are the two main crops with similar growing seasons (barley is also cultivated, to a much smaller extent) [53]. Acquiring images at two dates, pre-harvest and post-harvest, allow for maximum discrimination between opium poppy and other crops [55].

designed to produce estimates at a province level, and while district-level estimates are released, they are described in the following way: "They are indicative and suggest a possible distribution of the estimated provincial poppy area among the districts of a province" [57].

2.3 Socioeconomic Circumstances Surrounding Poppy Cultivation

Another body of related work tries to understand risk factors and consequences of poppy cultivation. In general, while opium poppy is seen to be a lucrative cash crop, the literature points to negative spillovers, such as food insecurity, corruption and violence, drug addiction, and poorer development-related outcomes such as access to electricity grids, healthcare facilities, and schools for girls [55]. One source of such evidence is UNODC surveys of village headmen. The Afghanistan Research and Evaluation Unit has also studied local conditions and specific phenomena surrounding poppy cultivation, using qualitative and survey-based methods, as well as high-resolution satellite imagery [e.g., 33–35]. Other work has used official poppy cultivation estimates (released by UNODC) at a district or province-level to estimate the causal relationship between poppy cultivation and terrorism, infrastructure, and conflict [30, 41, 65]. In terms of risk factors, [29] uses a theoretical framework to produce spatial maps of the risk of poppy cultivation in Afghanistan, but do not formalize the relationship to actual cultivation. We complement these approaches by demonstrating the ability to pair our high-resolution maps with other freely-available grid-level data sets, to gain insight on socioeconomic conditions at a much more granular level. To our knowledge, this is the first application in which opium cultivation maps have been combined with grid-level data for this purpose.

3 METHOD

Our analysis proceeds in several steps that are guided by knowledge of characteristics of the opium poppy crop, including growing seasons and spectral properties, summarized from related work (see Section 2) in Table 1. The classification approach relies on the premise that major crops have different crop cycles and unique spectral properties [28, 53, 54, 67]. First, we use information from vegetation indices to derive optimal satellite imagery acquisition dates. Then, with acquired imagery before and after crops are harvested, we use an unsupervised clustering approach on selected training regions, to separate the pixels into suitable clusters. We identify the cluster corresponding to opium poppy using known crop characteristics. Finally, we fit the fitted model to the entire study area to obtain poppy cultivation maps. Additionally, we combine these derived maps with other publicly-available grid-level data set to investigate the socioeconomic characteristics surrounding cultivation. A more detailed description of our data and methodology is as follows:

3.1 Study Area

Our study area is Afghanistan, which accounts for 70–90% of the world's supply of illicit opium. Afghanistan has a land area of 65.29 million hectares, of which an estimated 12% is agricultural land [4]. Our analysis focuses on these agricultural areas. Afghanistan is divided into 32 provinces and 398 districts, and 195 have recorded some opium poppy cultivation in the last 5 years with available district-level data (2017–2021). To evaluate the efficacy of our method, we focus on the top poppy-producing regions in Afghanistan, specifically districts belonging to the top quartile of districts with poppy cultivation (by acreage; Figure 1A). In 2019, 2020, and 2021, these corresponded to 91.4%, 93.0%, and 93.2% of total opium poppy cultivation, respectively.

We analyze three growing seasons, from 2019 to 2021. This data period was chosen since 2019 is the first year in which high-resolution surface reflectance imagery is available for Afghanistan (see Section 3.2), and 2021 is the last year in which aggregated district-level cultivation statistics, which we use to evaluate our results, are available. Our method is easily extendable to later years (see Section 5). We analyze the main growing season, which begins in the Winter and ends in Spring/Summer.

3.2 Remote Sensing Data

Sentinel-2 Optical Data. Sentinel-2 is an Earth observation mission operated by the European Space Agency. It systematically acquires optical imagery at high spatial resolution (10 m to 60 m) over land and coastal waters. Images contain 12 spectral bands: four bands (visible and NIR) at 10 m, six bands (red edge and SWIR) at 20 m, and two bands (atmospheric) at 60 m spatial resolution. It is a constellation with two satellites; the revisit frequency of each satellite is 10 days and the combined constellation revisit is 5 days. Compared to other earlier satellites such as Landsat 7 and 8, Sentinel-2 offers a higher resolution with shorter revisit periods. Data are made publicly available by the European Space Agency, and are easily accessible on Google Earth Engine (https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S2_SR_HARMONIZED), requiring only a browser and a straightforward application process to gain access. Level-2A orthorectified atmospherically corrected surface reflectance imagery are available from 2017, but are only available for Afghanistan from 2019. Optical satellite data were corrected for cloud cover using Sentinel-2: Cloud Probability data (https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S2_CLOUD_PROBABILITY). These data are released by the European Space Agency to accompany the Sentinel-2 imagery. Each 10 m pixel has an estimated probability that the pixel is cloudy. We filter for cloud and cloud shadow components using standard recommended procedures[3] that identify clouds using a threshold on the cloud probability, and nearby dark pixels using the near infrared (NIR) band.

MODIS Vegetation Indices. The normalized difference vegetation index (NDVI) is computed from the NIR and red spectral bands, as $NDVI = \frac{NIR-Red}{NIR+Red}$, and measures the amount of green vegetation. NDVI values are between -1 and 1, where values close to 1 indicate dense vegetation, values close to zero are urban areas, and negative values indicate clouds and snow. The Moderate Resolution Imaging Spectroradiometer (MODIS), launched by NASA in 1999, is a satellite-based sensor used for earth and climate measurements. The MODIS vegetation indices are a data product released at 250 m resolution at 16-day intervals. These are derived from composite products, and contain a representative NDVI value for the 16-day period, selected based on data quality and cloud cover [23]. Data are available on Google Earth Engine from 2000 to the present day (https://

[3] https://developers.google.com/earth-engine/tutorials/community/sentinel-2-s2cloudless. Several input parameters are required: we use `CLOUD_FILTER = 50`, `CLD_PRB_THRESH = 70`, `NIR_DRK_THRESH = 0.2`, `CLD_PRJ_DIST = 5`, and `BUFFER = 50`.

through visual inspection. For the former, we sum the number of pixels corresponding to the identified poppy cluster for each year, and convert this to the number of hectares at a district level. We compare these against the “ground truth” district-level estimates as released by the UNODC. To qualitatively verify our model predictions, we visually inspect the RGB bands of the Sentinel-2 imagery at the derived optimal dates in which imagery was acquired, to evaluate if they are consistent with our expectations. Needless to say, this manual verification method is limited to a small number of examples. We select these examples strategically, including examples in the training areas, other districts where aggregated estimates show a good correspondence, over-prediction, as well as under-prediction compared to UNODC estimates.

3.7 Combining with other Grid-Level Data Sets

We combine the generated opium poppy maps with auxiliary grid-level data sets using spatial joins, to gain insight to the socioeconomic conditions surrounding poppy cultivation. We use six relevant secondary data sets, involving travel time to urban center, healthcare accessibility, infant mortality, years of schooling, relative wealth index (RWI), and HDI. These data range from 250 meter to 10 kilometer resolution. To best match the time frame available for these auxiliary data sets, we use cultivation maps for 2019. For each grid cell in the secondary data set, we compute the proportion of pixels with poppy cultivation, and the proportion of pixels with other agriculture, where other agriculture corresponds to areas classified by the k-means algorithm as being in the two non-poppy clusters. Then, among all grid cells with any agriculture (poppy or others), we compare the distributions of the various socioeconomic variables for grid cells with any amount of poppy cultivation, and those with none, that is, only other crops.

A more detailed description of each data set is as follows: (1) Travel time (in minutes) to the nearest urban center [64] is estimated at 1 kilometer resolution, as of 2015. It is created as part of the Malaria Atlas Project, and is available at https://data.malariaatlas.org/maps. This can be interpreted as a measure of how remote a location is. (2) The healthcare accessibility data set, also created by the Malaria Atlas Project [63], contains predictions for land-based travel time (in minutes) to the nearest hospital or clinic at 1 kilometer resolution, as of 2019. It is available at https://developers.google.com/earth-engine/datasets/catalog/Oxford_MAP_accessibility_to_healthcare_2019. (3) For infant (under 1) mortality, predictions are available for the number of under-1 deaths per 1,000 live births, at a 5 kilometer resolution, yearly from 2000–2017 [14]. These are available at http://ghdx.healthdata.org/record/ihme-data/lmic-under5-mortality-rate-geospatial-estimates-2000--2017, and we use data from 2017. (4) Years of schooling for males and females are available at a 5 kilometer grid level, yearly for 2000–2017 [39], at http://ghdx.healthdata.org/record/ihme-data/lmic-education-geospatial-estimates-2000--2017. We use data for mean years of schooling among those aged 15–49 disaggregated by sex, in 2017. (5) Relative wealth estimates, as of approximately 2018, are available at a 2.4 kilometer resolution [15]. This is a measure of household asset wealth relative to others in the same country. (6) Global high resolution estimates of the United Nations HDI are available at a 10 kilometer grid level in 2019 [44], at https://www.mosaiks.org/hdi. The HDI is a multidimensional measure of development, incorporating information about not only income, but also education and health.

In addition, to provide background on the physical characteristics of poppy-growing regions versus areas with exclusively other agriculture, we characterize their geographic and climatic conditions using a similar methodology. Here, we use data on elevation, precipitation, and temperature. Elevation data, based on radar measurements and released by NASA [25], are available at 90 meter resolution at https://developers.google.com/earth-engine/datasets/catalog/CGIAR_SRTM90_V4. Monthly precipitation and temperature, derived from observations and regional every pixel.[4] Corresponding NDVI profiles for the three clusters in 2020 and 2021 are in Appendix Figure A.5, with similar results. In the 3 years, we identify clusters 0, 1, and 2, respectively, to be the clusters corresponding to poppy.

Additionally, we visually inspect the results in the model training region to verify that these clusters are correctly chosen. Example imagery from 2020 for Nad Ali and Zhari districts that are part of the training region are in the first two rows of Figure 4. In the first row, the visual characteristics described in Table 1 are obvious. In Panel A (pre-harvest) we see areas that are darker green, likely corresponding to wheat, and areas that are a brighter green, likely corresponding to poppy. In Panel B, we see that many of the brighter green areas are now beige, consistent with poppy fields being plowed after harvest. Many of the darker green areas are now a brown shade, indicating the ripening of wheat. Panel C highlights areas corresponding to cluster 0, which is the poppy cluster as identified using the NDVI profiles. In this case, we see many poppy fields being correctly classified as poppy. Many of the wheat areas, particularly on the right side of the image, are correctly not classified as poppy. However, a good number of wheat fields are also misclassified, for example, in the area just below reference point A, as well as to the top right of reference point B. This is likely because the wheat has ripened sufficiently to be mistaken as poppy.

Similar patterns can be observed in Zhari district in the second row—most of the poppy pixels are correctly classified, while wheat pixels above reference point A and to the left of reference point B are misclassified as poppy. In this district, we also note the presence of other crops beyond poppy and wheat. For example, there is a different crop to the top right of reference point C, that has not ripened in the pre-harvest period. In the post-harvest period, it is much greener and closer to ripening. This crop has a different crop cycle, and the careful selection of image acquisition dates results in these pixels correctly not being classified as poppy. This point is true more generally in other districts with different crop mixes: the selection of image acquisition dates maximizes the likelihood of there being the two main crops of wheat and poppy acquired at the correct stages of the growing cycle, particularly in the regions of interest with larger amounts of poppy cultivation. This validates our decision to focus on the two main crops of poppy and wheat, and justifies the selection of $k = 3$ for the number of clusters.

4.2 Comparing with Aggregate Statistics

Figure 3 plots the aggregated district-level acreage estimates from our method against UNODC estimates for 2019, 2020, and 2021, respectively, for districts in the top quartile of cultivation in each year. We see a strong correlation between our estimates and UNODC estimates, with a Pearson correlation correlation of 0.777, 0.758, and 0.814 for 2019, 2020, and 2021, respectively. For all three years combined, the correlation coefficient is 0.771.[5] We see some over-prediction, particularly for the two top-producing districts, which we investigate further through visual examination in Section 4.3.

In these figures, we note that model performance is much more variable in districts with lower levels of cultivation (leftmost data points in the figures). This is likely because in these districts, the crop mixes are much more heterogeneous, with opium poppy being only a minor crop. Computed optimal acquisition dates may correspond to other major crops that have a different growing cycle than opium poppy, and so imagery acquired may not capture opium poppy at the right stages of

[4] As a secondary validation of the results, we produce a standard t-SNE plot [59] showing the separation between clusters (Appendix Figure A.4).

[5] $R^2 = 0.59$. For comparison, the prediction accuracy (measured using R^2) for other sustainable development outcomes using satellite imagery and machine learning are around 0.65 for village/district-level asset wealth, and 0.42 for plot-level smallholder yields [13].

center, we find that compared to agricultural areas with no poppy cultivation, grid cells with any amount of poppy cultivation have shorter travel times to urban areas (mean of 193 minutes versus 237 minutes for other agricultural areas). Related to this, poppy-growing grid cells have on average better access (shorter travel times) to healthcare facilities (mean of 89 minutes versus 225 minutes for other agricultural areas). On the other hand, poppy-growing areas have higher infant mortality (mean of 2.6 under-1 deaths per 1,000 versus 1.50 for other agricultural areas) and lower years of schooling (males: mean of 5.47 years for poppy-growing areas versus 7.55 for other agricultural areas; females: 2.08 years versus 2.80 for other agricultural areas). Boxplots showing the distribution of these variables are in Figure 6. Finally, grid cells with any amount of poppy cultivation have slightly lower scores both on the RWI (mean of -0.46 for poppy-growing versus -0.40 for non-poppy agricultural areas) and the HDI (mean of 0.38 for poppy-growing versus 0.40 for non-poppy agricultural areas) (Figure 7). Statistical tests for differences in means between the no poppy and any poppy groups are significant in all cases ($P < 0.001$).

In general, the literature points to poppy-cultivating areas as having more insecurity and less government presence, lower levels of development, and associated indicators such as access to schools and healthcare facilities [29]. Our results are largely consistent with these predictions. The finding that areas with poppy cultivation have shorter travel times to urban centers is interesting and merits further discussion. There are conflicting accounts from the literature: [29] suggests that lack of accessibility contributes to socio-economic vulnerability, which is a risk factor for opium poppy cultivation. Yet [65] finds that infrastructure development in the form of road construction leads to more opium poppy cultivation. Here, our results are consistent with the latter argument. The result on accessibility to healthcare follows from this result: accessibility to healthcare is a function of two quantities: transportation infrastructure and the availability of healthcare. We might expect poorer healthcare availability in poppy-growing regions, consistent with more insecurity and a lower level of development—yet likely because of the better overall transportation infrastructure, we observe shorter travel times to healthcare facilities in poppy-growing areas.

Another interesting finding is that the shorter travel time to a healthcare facility does not translate to lower infant mortality (Figure 6B). This could be because of numerous other barriers to receiving care, such as poverty, low literacy rates, lack of female healthcare providers, insecure travel, and an inability of women to travel without a male escort [6]. These factors might be exacerbated in opium-cultivating areas; for example, constraints on the movement of women are particularly acute in Taliban-controlled, opium-cultivating areas [36]. As a result, lower travel times do not necessarily translate to women and infants receiving appropriate care. One important caveat, however, is that while the other auxiliary data sets used in Figure 6 are geographically complete, the infant mortality data set is missing estimates in more rural areas due to availability of data; this results in about 15% of agricultural pixels being dropped from the analysis. Additionally, for the mortality and education outcomes, we use the most recent year of data available, 2017, paired with opium poppy predictions for 2019. While more recent data are not available, these socioeconomic outcomes tend to change slowly across years.

In terms of development indices, poppy-growing areas score slightly worse on two such indices, RWI, and HDI. The RWI data set predicts asset-based relative household wealth using high-resolution satellite imagery, data from mobile phone networks, topographic maps, and Facebook connectivity data. The HDI data set predicts HDI as defined by the United Nations Development Program (UNDP)—an average of achievements in health, education, and standard of living [48]—using satellite imagery. While there are limitations associated with both these data sets, in particular prediction errors and geographical incompleteness due to data availability, they point to the same conclusion, which is that poppy-growing areas have lower levels of wealth and

B APPENDIX B: REPRODUCIBILITY

Our analysis was carried out in a compute environment composed of two 12 core Intel Xeon Silver 4214R 2.4 G CPUs with 768 GB RAM. This is not a requirement but an artifact of our implementation. Note that somebody interested in running the study end-to-end might be required to adjust some of the parameters—including grid-cell resolution, number of CPU cores to be used and so on. – based on the specifications of the environment in which this code is run. Required parameters are fully described and documented.

All of the code used for this analysis is made publicly available at https://github.com/arogyakoirala/poppy. This contains roughly 1,700 lines of Python (Python 3.9.7) code that have been extensively documented for public dissemination. No non-standard packages are used, and all required packages and their versions are documented in a `requirements.txt` file.

Step-by-step instructions for scripts that execute the four stages required to generate the maps – data preparation, download, model fitting, and prediction—along with descriptions of required arguments have also been documented. All of the data inputs required for the analysis, with the exception of the Sentinel-2 imagery, are set up in the data preparation step. This includes ground truth cultivation statistics from UNODC, a district-level geopackaged shapefile for Afghanistan, geopackaged shapefiles for modelling regions within Zhari and Nad Ali, and the Copernicus Global Land Cover crop mask.

Sentinel-2 imagery from Google Earth Engine is downloaded separately using the `download.sh` script in the data download step, and image extraction and processing is carried out as described in Section 3.3. We reverse the order of operations in the implementation to allow parallelization of the imagery download (at the 50 km grid level), while preserving the same results. To be clear, in the implementation we first generate 50 km grid cells, and within each 50 km grid cell, acquire Sentinel-2 imagery for every qualifying 250 m tile, from Google Earth Engine.

With the downloaded and processed data, the model is fit and maps for all districts that we analyze are produced in the fitting and prediction stage (`fit.py` and `predict.py`). Rasters of cultivation maps are available as Google Earth Engine assets. For the socioeconomic analysis, poppy cultivation rasters were combined with other raster data sets on Google Earth Engine. Links to Earth Engine code are provided in the GitHub repository.

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